

NAME: _____

UIN: _____

QUIZ 1: 2 problems (10 pts)**Closed book/Closed notes: Show all your work for full credit**

1. (6 pts) The average atmospheric pressure on Earth is approximated as a function of altitude by the relation $P_{\text{atm}} = 101.325(1 - 0.02256z)^{5.256}$, where P_{atm} is atmospheric pressure in kPa and z is altitude in km with $z=0$ at sea level. A monometer is used to measure the pressure of a gas in a tank in Denver ($z=1610\text{m}$). The fluid has a density of 850kg/m^3 and the monometer column height is 200cm . Determine the absolute and gage pressure of gas within the tank.

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2. (4 pts) A gas contained within a piston-cylinder assembly undergoes an expansion process for which the relationship between pressure and volume is given by $PV^n = \text{constant}$

The initial pressure is 2 bar, $V_1 = 2.5 \text{ m}^3$ to $V_2 = 4 \text{ m}^3$. Determine the work for the process in kJ, if

- a. $n = 0$ and
- b. $n = 1.5$
- c. Sketch both processes on a P-V diagram